

15. Consider the following recursive procedure. Assuming that n is a power of 2, what is the asymptotic time complexity of $F(n)$?

procedure $F(n)$:

if $n \leq 1$:
 return
end if

$O(n \log n) = O(\log 6)$

for $i = 1$ to n :

$j = 1$

 while $j \leq n$:

 perform one constant-time operation

$j = 2 * j$

 end while

end for

$F(n/2) = F(n/4) \rightarrow \log n$

$F(n/2)$

$n \cdot \log n$

- (A) $\Theta(n)$
- (B) $\Theta(n \log n)$
- (C) $\Theta(n^2)$
- (D) $\Theta(\log^2 n)$
- (E) $\Theta(n \log^2 n)$

16. Given the table Orders(customer, amount):

	customer	amount	
①	Ann	60	110
⑤	Bob	120	
	Ann	50	130
②	Cara	70	
	Bob	10	110
	Dora	200	
	Cara	40	

What is the result of executing the following SQL query?

```
SELECT customer
FROM Orders
GROUP BY customer
HAVING COUNT(*) >= 2 AND SUM(amount) > 100
ORDER BY customer;
```

- (A) Ann, Bob
- (B) Bob, Cara
- (C) Ann, Bob, Cara
- (D) Ann, Cara, Dora
- (E) Dora

注意:背面有試題